1)

The Mafia board game was invented in 1986 by Dmitry Davydov, a student at Moscow State University. The plot of the game is simple: civilians of the city want to find the Mafia, while the Mafia wants to kill everyone in the city except Mafia members.

In the classic variation of the game, there are 12 characters in the game: 1 Don (Mafia Leader), 2 Mafia, 1 Doctor, 1 Commissar and 7 Civilians. Characters are selected by drawing cards with roles.

Oliver, Martin and Jacob play Mafia with their friends. There are 12 players in total. What are the chances that at least two of them will be civilians?

A. 840/1320
B. 420/1320
C. 480/1320
D. 630/1320

2)

Exclusive or (XOR) is a logical operation that compares two binary digits and returns a 1 if one of the inputs is 1 and the other input is 0. It is represented by the symbol "⊕" or "^". The truth table for the XOR operation is as follows:

A	B	A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

A bitwise XOR operation compares each bit of one binary number to the corresponding bit of another binary number. For example, if we have two bit numbers A and B, A^B is a new binary number C where each bit of C is the result of the XOR operation on the corresponding bits of A and B. Here is an example:

bitwise xor

```
  01010001110101110000000000001111
^ 00110001011011100011000101101110
  01100000101110010011000101100001
```

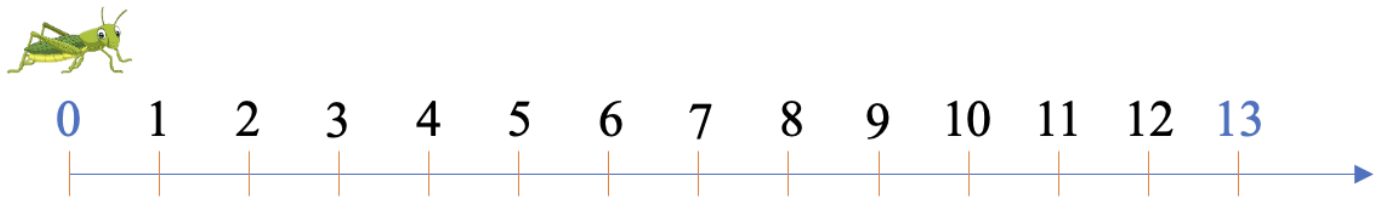
By first converting into binary numbers, what is the result of "13^8"? (Remember this is not 13 to the power of 8!)

A. 5
B. 13
C. 104
D. 21

3)

Peter caught a grasshopper in his garden. After a while, he noticed that it can only jump at fixed distances, so he decided to play a game with it.

Peter put the grasshopper along a ruler at the 0cm mark. The grasshopper only jumps forward, and travels either 1cm or 2cm every jump.



How many distinct combinations of 1cm jumps and 2cm jumps are there for the grasshopper reach the 13cm mark?

A. 377
B. 233
C. 7
D. 144

4)

The StemCoTech company is a business that sells laptops.

One day, an accident occurred, resulting in 30 percent of the laptops in a batch becoming defective. Before the StemCoTech company became aware of this incident, another company, MidTech, purchased 10 laptops from this defective batch.

Of the 10 laptops they purchased, MidTech only needs 8 or more functioning laptops, as the company has 8 employees currently.

What is the probability that MidTech will have enough working laptops to start its operations?

A. 38%
B. 62%
C. 23%
D. 5%

5)

There are different types of bank deposit systems, with the most popular being the annual deposit system. In this system, a certain percentage is added to the account holder's balance once a year. On the other hand, the semi-annual system involves the addition of half that specific percentage to the account balance twice a year (ie every six months).

For example, if someone deposits $100 into an annual deposit account with a 5% annual interest rate, they will have $105 in their account at the end of the year, $110.25 at the end of two years, and so on. In contrast, if they deposit $100 into a semi-annual account with a 5% annual interest rate, they will receive 2.5% every six months, resulting in a balance of $102.5 after six months and $105.0625 after one year, and so on.

Mike invested $10000 in two different banks, one with an annual interest rate of 10% per year, and the other with a semi-annual interest rate of 10% per year.

What will be the difference in the account balances between these two banks after 5 years?

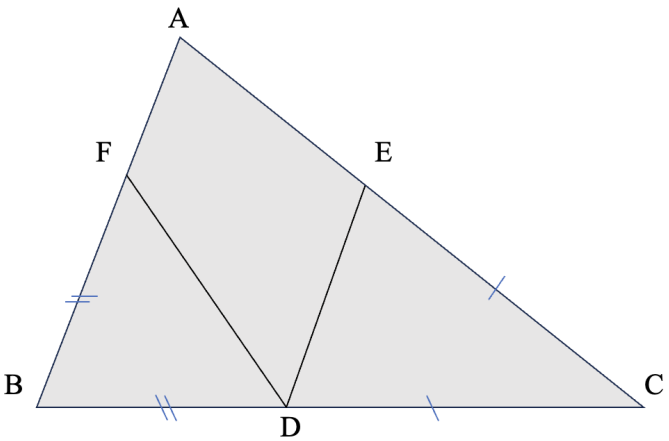
A. $184
B. $0
C. $213
D. $281

6)

Aibek and Maksat are seated at the same table during their math lesson. Their geometry teacher assigned them a simple task: draw a triangle and measure one of its angles using a protractor.

Unfortunately, Maksat forgot his protractor at home and had to ask Aibek for assistance in measuring the angle. Aibek, being a good friend, was willing to help Maksat, but he also saw an opportunity to teach him a valuable lesson.

Instead of simply measuring one angle, Aibek decided to add three more points to the triangle, labeling them as D, E, and F. He positioned these points in a way such that BF was equal to DB and DC was equal to EC:



Following this, he measured the angle ∠FDE to be 50 degrees and confidently stated that he could use this information to determine the measure of angle ∠BAC. What is the value of this angle?

A. 80°
B. 50°
C. 130°
D. 70°

7) Under the tenure of health minister Mr. Ong Ye Kung, Singapore rolled out the mandatory Nutrigrade classification of all drinks, which assigns a grade level to drinks based on their sugar and fat content.

Let's say that a (fictitious) bubble tea chain, Hecha Sichen, rolled out a new bubble tea drink which was declared to have 5 g of sugar per cup. Mr. Ong was skeptical of that claim, and hence he decided to purchase 100 cups of that drink, and analyzed the sugar content of this sample of 100 drinks. From his sample, he found that the mean sugar content in a cup is 6.2 g, with the sample standard deviation being 4.8 g. Mr. Ong then decided to perform a hypothesis test on the hypothesis that this bubble tea drink actually contains **more** than 5 g of sugar per cup on average. What would be the p-value of such a test? (You may assume that the t distribution with n degrees of freedom can be approximated as a normal distribution if n>30.)

A. 0.0062
B. 0.0431
C. 0.0122
D. 0.0075

8) Three friends, Ian, Zhi Jie, and Yong Boon, saw that the Ippudo ramen restaurant was offering a 1-for-1 promotional sale. They hence decided to go for this good deal.

Zhi Jie has a project meeting on that day, hence there is a probability p that he is unable to join his friends. In that case, only Ian and Yong Boon will head to the restaurant. If all three friends meet at the restaurant, there is a probability q that they will buy 4 bowls of ramen, and a probability 1 - q that they will only buy 2 bowls of ramen. If only two of them are present, they will only buy 2 bowls of ramen. Given that 2 bowls of ramen was ordered that day, what is the probability that all three friends turned up at the restaurant?

A. (1 - q)(1 - p)/(1 + pq - q)
B. (1 - q)/(1 - p)
C. (1 - q)/(2 - p)
D. pq/(pq + 1 - p)

9) A group of 10 friends decided to sign up for the StemCo Mathematics competition to best themseleves. They were then informed that each of them has a chance to qualify for the StemCo Global tournament in Singapore if they scored sufficiently well, a great opportunity for them to learn more about STEM and cultivate a healthy "Best Yourself!" mentality.

Suppose that each person in the group of 10 friends has an equal probability of 0.4 of qualifying for StemCo Global, and the event that each person qualifies for the tournament is independent of the results of the other friends. Let X be the random variable representing the number of people in the 10 friends who manage to qualify for StemCo Global. What is the standard deviation of X?

A. 1.55
B. 2.40

C. 4
D. 2

10) Shinsaibashi is one of Osaka's most famous shopping streets. It is truly a shopaholic's paradise with famous brands, uniquely Japanese souvenirs, and lots of delicious street food.

On one busy Saturday evening, the rate of people people entering Shinsaibashi from entrances A and B, t hours after 5pm, are modelled by the equations: r_A = 500t + 120 people/hour r_B = 300t + 230 people/hour The rate of people leaving Shinsaibashi from the only exit is given by r_E = 900t people/hour Assume that there are zero people in Shinsaibashi at 5pm, and that Shinsaibashi closes at 10pm. What is the maximum number of people in Shinsaibashi on this Saturday evening?

A. 613
B. 697
C. 352
D. 499

11) Nara Park is a popular park in Nara, Japan, known for the friendly herds of deer roaming the park. It is a common tourist activity to buy snacks to feed the deer and interact with them.

A statistician is modelling the population of deer in Nara Park, with the number of deer denoted by the variable N, and the variable t referring to the number of years after 2023. The rate of births of deers is modelled by the equation: Rate of births = 0.4*N deer/year The rate of deaths of deers is modelled by the equation: Rate of deaths = 0.2*N deer/year Additionally, the park managers will often catch aggressive deer and send them to a nature park far away from visitors, at an average rate of 5 deer/year. If the initial population of deer is 300, how many deers will there be by the end of the year 2030?

A. 1140
B. 1130
C. 1060
D. 1038

12) An epidemologist was studying an influenza outbreak in a city.

He modelled the number of people infected with the disease (y) over time in days (x, where x>0).

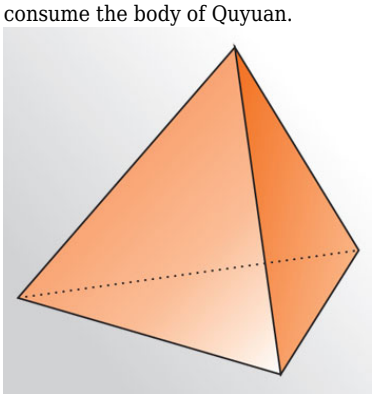
$$y = \frac{500}{1 + e^{-0.01(x-300)}}$$

On which day is the number of infected people increasing at the highest rate?

A. 300
B. 623
C. 214
D. 600

13) In Chinese tradition, Quyuan was a righteous officer of the state of Chu in China. He was, however, sidelined by the corrupt King, whose mismanagement of the country eventually resulted in the fall of the state of Chu.

Quyuan was filled with so much sorrow at the demise of his country that he drowned himself in the Miluo river. Later, it became a tradition to make rice dumplings in the shape of tetrahedrons (example below), and throw them into the river for the fish to eat, such that they would not consume the body of Quyuan.



Tetrahedral dumplings are still consumed by the Chinese people till this day on Duanwu Festival to remember the story of Quyuan. Jane wanted to make tetrahedral dumplings with the faces of the tetrahedron being equilateral triangles with a length x in centimeters. Let the volume of the dumpling be V cm^3 and the surface area be A cm^2. Given that V - A = 10x, what is the value of x?

A. 19.1
B. 15.5
C. 19.6
D. 4.3

14) In Chinese tradition, after Cao Pi ascended the throne of the Kingdom of Wei, he felt threatened by his intelligent younger brother, Cao Zhi, and wanted to find a way to get rid of him. Cao Pi challenged Cao Zhi to compose a poem within the timeframe of walking 7 steps. Amazingly, the intelligent Cao Zhi managed to compose a poem which impressed his brother, thus saving his life. This poem is now known as

The Seven Steps Poem, learnt by many Chinese students till this day.

七步诗

——[三国.魏]　曹植

煮豆持作羹，
漉菽以为汁。
其在釜下燃，
豆在釜中泣。
本自同根生，
相煎何太急？

Suppose that the 7 steps taken by Cao Zhi covered a total distance of 150 cm, with the distance of each step following a geometric series, with the first step covering a distance of a cm and the next step being 0.9 times the length of the step before it. If Cao Pi were to ask Cao Zhi to walk 100 steps instead, what would be the distance covered?

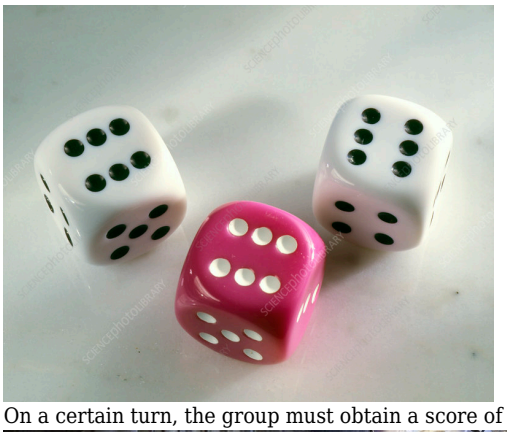
A. 287.5 cm
B. 214.3 cm
C. 159.5 cm
D. 167.3 cm

15) In 209 AD, the Battle of Chi Bi was recorded by Chinese historians to have occurred between the general Cao Cao and his enemies. Cao Cao brought a huge fleet of ships to the battle, but the intelligent advisor of his rivals, Zhu Geliang, vanquished Cao Cao's fleet. This was because Cao Cao chained all his ships together to prevent them from swaying in the current. Zhu Geliang creatively started a fire on one ship, which quickly spread to the other ships because they were chained close to one another.

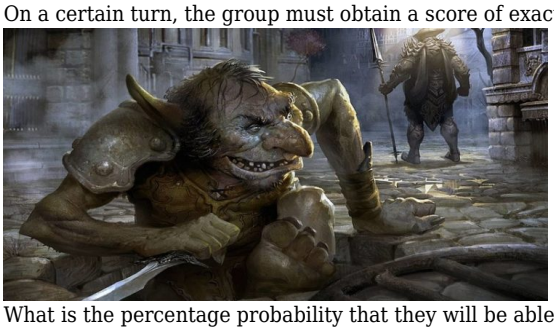
Suppose that the fire spread from one ship to another, one at a time, following an arithmetic progression. It took 10 minutes for the fire to consume the first ship, and for each successive ship, it took 5 minutes more for the fire to consume that ship than the time taken for the previous ship. If Cao Cao had 100 ships, how long did it take for all of his ships to be consumed by fire?

A. 25 750 minutes
B. 43 135 minutes
C. 15 596 minutes
D. 19 695 minutes

16) A group of friends got together to play a fantasy board game. During each turn of the game, they will need to roll 3 dice to decide their game progress. The total score from their dice roll is determined by multiplying the scores on each of the 3 dice.



On a certain turn, the group must obtain a score of exactly 24 in order to defeat a pack of goblins that lie ahead.



What is the percentage probability that they will be able to defeat the goblins (rounded to the nearest whole number)?

A. 7%
B. 8%
C. 9%
D. 10%

17) Given a set of data, statisticians have many different ways of generating predictive trends. The most common method is to draw a line of best fit, which predicts linear growth. While useful, the method has a disadvantage - the line does not have to necessarily pass through each data point exactly! As such, other methods have been devised. One such method is called the "Lagrange Interpolation". Given a set of data points

$$(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$$

, where none of the x-coordinates are the same, this method generates a "Lagrange Interpolating Polynomial", which passes through all of these points. Moreover, this polynomial is the smallest-degree polynomial that satisfies this. The Lagrange interpolating polynomial is given by the following formula:

$$f(x) = \sum_{j=1}^{n} \left[y_j \left(\prod_{1 \le i \le n,\ i \ne j} \frac{x - x_i}{x_j - x_i} \right) \right]$$

Given data points (-2, 5), (-1, 2), (0,-3) (1, -4), what is the value of f(2), where f(x) is the Lagrange Interpolating Polynomial through these four data points?

A. 5
B. 8
C. -5
D. 0

18) Three-dimensional real vectors can undergo two different types of products - the dot product a . b, which produces a scalar, and the cross product a x b, which produces another three dimensional vector. Jack is now given **x**, **y**, and **z**, where each are a random three-dimensional real vector. How many of these quantities has to be 0 (irregardless of the choice of vectors)? (i)

$$(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{x}$$

(ii)

$$|(\mathbf{x} \times \mathbf{y}) \times \mathbf{x}|$$

(iii)

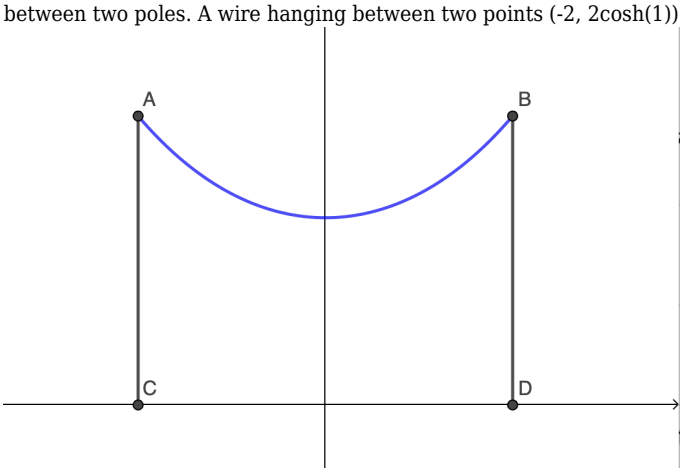
$$(\mathbf{x} \times \mathbf{y}) \cdot (\mathbf{x} \times \mathbf{z})$$

(iv)

$$|(\mathbf{x} \times \mathbf{y}) \times (\mathbf{x} \times \mathbf{z})|$$

A. 1
B. 0
C. 2
D. 3

19) A caternary is a curve with the equation y = a cosh(x/a), where cosh is the hyperbolic cosine function given by cosh(x) = $(e^x+e^{-x})/2$. It is mainly used to describe the shape of a loose cable that hangs inbetween two points of the same height, such as electrical wires hanging between two poles. A wire hanging between two points (-2, 2cosh(1)) and (2, 2cosh(1)) has the equation of a catenary y = 2 cosh(x/2).



Let s be the length of the wire. Which of the following is correct? Hint: The arc-length s of a function y from x = a to x = b is given by

$$s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

.

A. 4.5 ≤ s < 5
B. 4≤ s < 4.5
C. 5≤ s < 5.5
D. 5.5 ≤ s < 6

20) In physics class, Annie learnt that the jerk of an object is defined as the rate of change of acceleration over time. She also knows that acceleration is the rate of change of velocity over time, and velocity is the rate of change of displacement over time. Annie sees some children playing in a playground, and she wants to apply her knowledge and calculate the jerk of the swing in the playground.

Annie observes that the swing's displacement from the center in meters is 4cos(2t), where t is time in seconds and cos is in radians. What is the swing's jerk at the 10 second mark?

A. 29.2 ms^{-3}
B. -29.2 ms^{-3}
C. 10.9 ms^{-3}
D. -10.9 ms^{-3}

21) Earthquakes are powerful natural disasters that are measured by their magnitude on seismographs.

Each increase in magnitude by 1 means the earthquake releases 32 times more energy. For instance, a magnitude 5 earthquake has 32 times more energy than a magnitude 4 one. The strongest earthquake ever recorded was the 1960 Valdivia earthquake in Chile, with a magnitude of 9.5. In 2001, the Geiyo earthquake in Japan had a magnitude of about 6.7. How many times more energy does the Valdivia earthquake release compared to the Geiyo earthquake?

A. 16384
B. 89.6
C. 28
D. 8192

22) The volume of water in a hemispherical bowl of radius 12cm is given by $V = (\pi/3)[36x^2 - x^3]$, where x is the depth of the water.

Water is poured in at the constant rate of 4 cm^3s^{-1} and leaks out from a hole at the bottom of the bowl at a rate of k cm^3s^{-1}. Given that the water level is rising at a rate of $1/(21\pi)$ cm s^{-1} when the depth is 3cm, find the value of k.

A. 1
B. 1.5
C. 2
D. 3

23) Emmy had a house party at her home and made 7 different types of snacks. Then, five of Emmy's friends came to the party and tried every type of snack.

After the party, Emmy's five friends told her what their favourite type of snack was. Assuming each snack has the same chance of being picked as favourite, what is the probability that at least 2 people have the same type of snack as their favourite?

A. 85%
B. 71%
C. 96%
D. 63%

24) Harold Houdyni is performing a magic show in Las Vegas tonight.

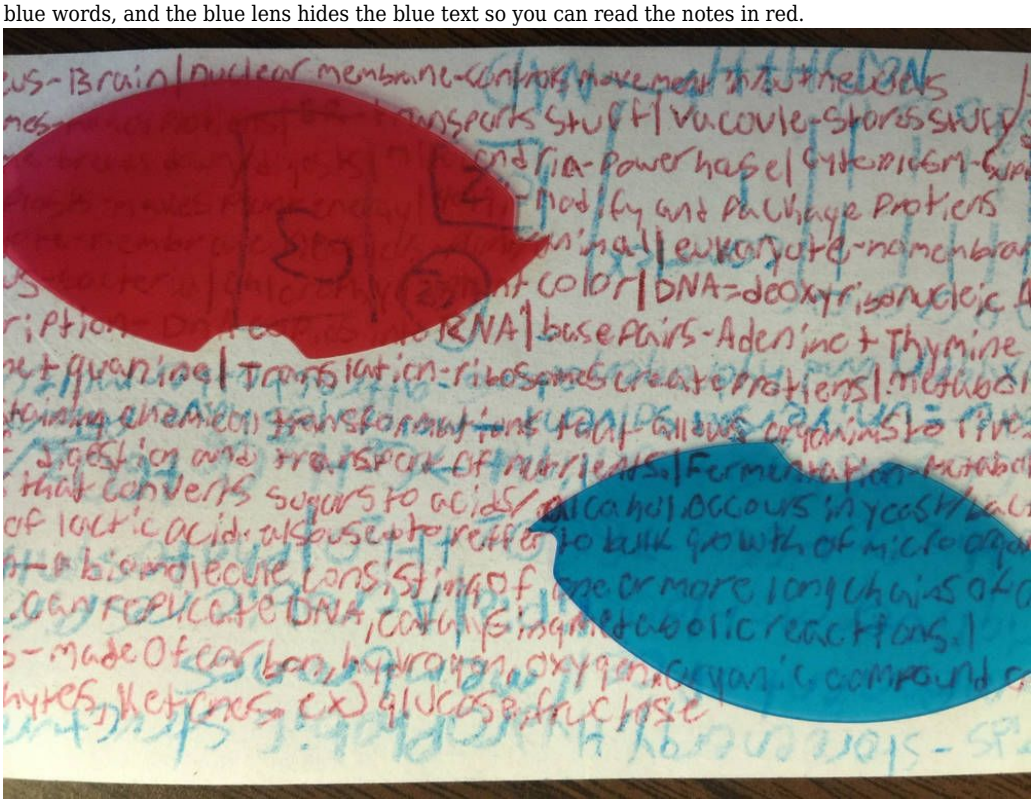
By sheer luck, you have been picked by him to be his guest on stage. He shows you four hats and places a ball underneath one of the hats. He proceeds to shuffle the hats, but your vision is unable to keep up with his dexterity, and you lose track of the hat with the ball. When he asks you which of the hats has the ball, you decide to randomly pick a hat. Before Houdyni lifts up your chosen hat, he lifts up another hat that does not have the ball. He then gives you a chance to change your answer. If you were to change your answer to one of the other two hats, what would be your percentage chance of choosing the hat with the ball?

A. 37.5%
B. 25%
C. 75%
D. 33.3%

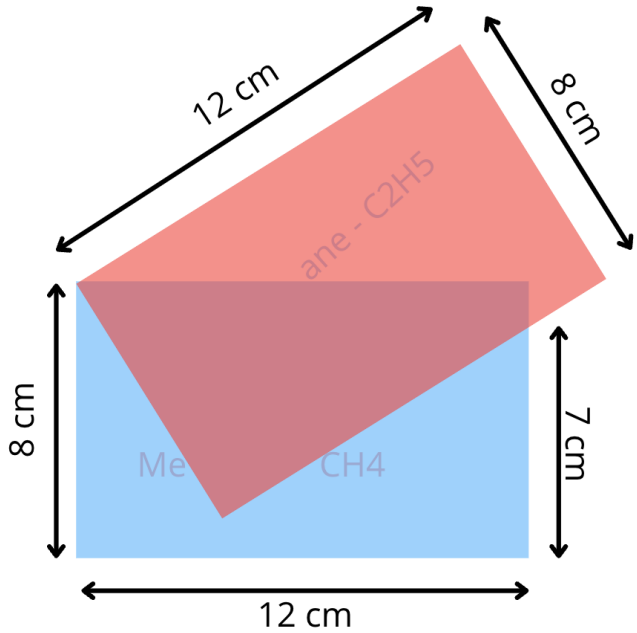
25)

A cheatsheet is a single-page document containing condensed and relevant information that students are allowed to bring into the exam by some professors. It serves as a personalized reference guide, enabling students to have key concepts, formulas, or notes at their fingertips to aid them during the test, without the need to memorize everything.

Some students find one page to be too small to contain all of the necessary information, which is why they use red and blue coloured pens and two colored lens that allow them to double the amount of information one page can contain. The red lens hides the red text so you can read the blue words, and the blue lens hides the blue text so you can read the notes in red.



Mark uses two rectangular lens during the exam and at one moment the lens overlapped. Mark found out that there is no text in the intersection of the two lens:



What is the area of this intersection region?

A. 42 cm^2
B. 96 cm^2
C. 36 cm^2
D. 48 cm^2

26)

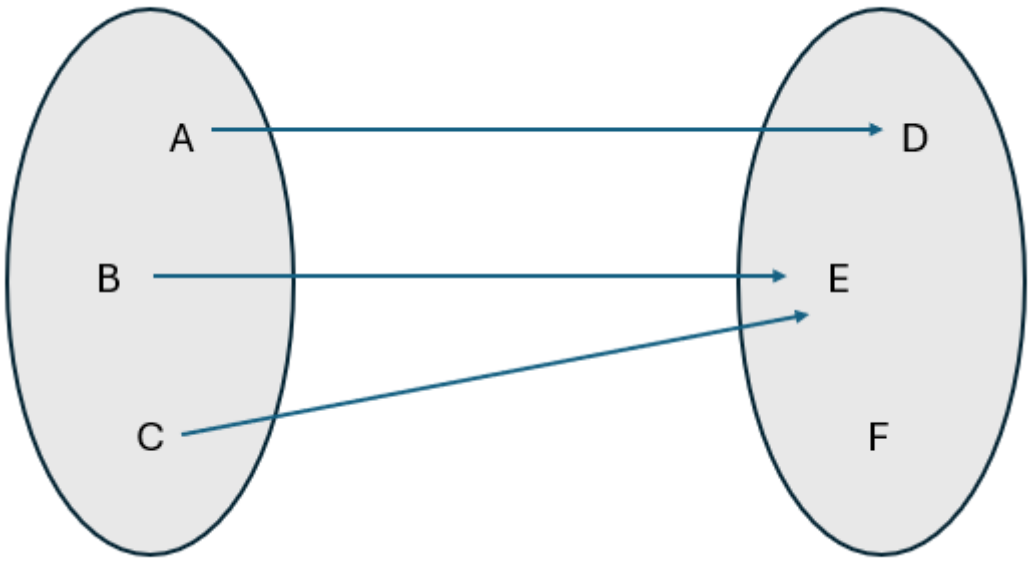
On John's shelf, there are books written in different languages. Michael visits John's house, and feeling bored, he randomly draws two books from the shelf.

It is known that the probability of drawing two books in Spanish is 5/14.

Which of the following can be the number of books on John's bookshelf?

A. 8
B. 7
C. 9
D. 5

27) A function f(x) can be defined as a mapping between elements of an input and elements of an output. We call a function one-to-one if every possible output of the function corresponds to one unique input. Different outputs should correspond to different inputs.



For example, this function here is not one-to-one as E can correspond to both B and C. But if C->F instead, then it would be one-to-one. As another example, the function $f(x) = x^2$ that accepts and outputs real numbers is not one-to-one, since the output 4 could correspond to the two inputs 2 and -2. Assume that each of the following functions accepts any real number input. Which of the following functions is one-to-one?

A. $f(x) = x^3$
B. $f(x) = \sin x$
C. $f(x) = |x + 1|$
D. $f(x) = x^6$

28) Mr Wan is a lazy teacher and does not want to teach his class. During his one-hour math lesson this week, he tells his students to calculate the last three digits of 5^{50}, thinking that they would spend the entire hour trying to compute the value of this impossibly large number, so that he would get to relax. But to his shock, the bright student Ted noticed a pattern in the powers of 5, and managed to give him the correct answer in less than a minute! What was Ted's answer?

A. 625
B. 125
C. 025
D. 375

29) Alicia needs to take long-term medication for a medical condition. She takes three different types of pills, a blue pill, a pink pill, and a white pill.

She takes 2 blue pills every 3 days, half of a pink pill every 2 days, and one-third of a white pill every day. When she has run out of a certain type of pill, she will visit the hospital on the day she is next supposed to take the pill, and she will take it on the same day after her visit. She always collects 20 of each pill every time. Her doctor gives her 20 of each pill on 6 October 2023 and she starts taking all her medication that day. What is the month where she next collects all three types of pills from the hospital on the same day?

A. June
B. May
C. March
D. July

30) Kenneth is in China visiting the famous Stairway To Heaven, which is a flight of 999 steps.

He wants to climb to the top, but he can only cover a maximum of 4 stairs in each step. Let a_k represent the number of ways he can reach the top of a flight of k stairs with only 1, 2, 3, or 4 stairs in each step. He wants to find the value of a_{999}. If he already knows the values of a_{998}, a_{997}, a_{996}, and a_{995}, which of the following statements is true?

A. $a_{999} = a_{998} + a_{997} + a_{996} + a_{995}$
B. $a_{999} = a_{998} + 2a_{997} + 3a_{996} + 4a_{995}$
C. $a_{999} = 4a_{998} + 3a_{997} + 2a_{996} + a_{995}$
D. $a_{999} = a_{998} + 2a_{997} + 4a_{996} + 8a_{995}$